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Permutations and Combinations

JEE Main Mathematics · Previous Year Questions with Answer Key

233	179	2002 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

INSIDE THIS PDF

MCQ (Single Correct Answer)	139 questions
Numerical	94 questions
Answer key	at the end

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MCQ (Single Correct Answer) 139 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

A person has three different bags and four different books. The number of ways, in which he can put these books in the bags so that no bag is empty, is :

(A) 18 (B) 36 (C) 39 (D) 72

Open solution

Q2. JEE Main 2026 (Online) 6th April Evening Shift

A building has ground floor and 10 more floors. Nine persons enter in a lift at the ground floor. The lift goes up to the 10th floor. The number of ways, in which any 4 persons exit at a floor and the remaining 5 persons exit at a different floor, if the lift does not stop at the first and the second floors, is equal to :

(A) 2184 (B) 3064 (C) 7056 (D) 11340

Open solution

Q3. JEE Main 2026 (Online) 6th April Morning Shift

The number of 4-letter words, with or without meaning, each consisting of two vowels and two consonants that can be formed from the letters of the word INCONSEQUENTIAL, without repeating any letter, is:

(A) 2670 (B) 2840 (C) 2920 (D) 3600

Open solution

Q4. JEE Main 2026 (Online) 5th April Evening Shift

A box contains 5 blue, 6 yellow and 4 red balls. The number of ways, of drawing 8 balls containing at least two balls of each colour, is :

(A) 4100 (B) 4140 (C) 4230 (D) 4290

Open solution

Q5. JEE Main 2026 (Online) 4th April Evening Shift

Let

$\mathrm{A} = \{(a, b, c) : a, b, c$ are non-negative integers and $a + b + 2c = 22\}$.

Then $n(\mathrm{A})$ is equal to :

(A) 121 (B) 124 (C) 144 (D) 169

Open solution

Q6. JEE Main 2026 (Online) 4th April Morning Shift

The number of ways, of forming a queue of 4 boys and 3 girls such that all the girls are not together, is:

(A) 5040 (B) 3050 (C) 3410 (D) 4320

Open solution

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Q7. JEE Main 2026 (Online) 2nd April Evening Shift

Let p_n denote the total number of triangles formed by joining the vertices of an n-side regular polygon.
If $p_{n+1} - p_n = 66$, then the sum of all distinct prime divisors of n is :

(A) 7 (B) 8 (C) 5 (D) 6

Open solution

Q8. JEE Main 2026 (Online) 2nd April Morning Shift

The number of seven-digit numbers, that can be formed by using the digits 1, 2, 3, 5 and 7 such that each digit is used at least once, is:

(A) 15400 (B) 17800 (C) 16800 (D) 29400

Open solution

Q9. JEE Main 2026 (Online) 2nd April Morning Shift

The number of elements in the set $S = \left\{(r, k) : k \in \mathbb{Z} \text{ and } {}^{36}C_{r+1} = \frac{6\left({}^{35}C_r\right)}{(k^2-3)}\right\}$ is :

(A) 2 (B) 4 (C) 8 (D) 16

Open solution

Q10. JEE Main 2026 (Online) 28th January Morning Shift

Let $\mathrm{S} = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$. Let x be the number of 9-digit numbers formed using the digits of the set S such that only one digit is repeated and it is repeated exactly twice. Let y be the number of 9 - digit numbers formed using the digits of the set S such that only two digits are repeated and each of these is repeated exactly twice. Then,

(A) $56x = 9y$ (B) $21x = 4y$ (C) $45x = 7y$ (D) $29x = 5y$

Open solution

Q11. JEE Main 2026 (Online) 24th January Evening Shift

The letters of the word "UDAYPUR" are written in all possible ways with or without meaning and these words are arranged as in a dictionary. The rank of the word "UDAYPUR" is

(A) 1579 (B) 1578 (C) 1580 (D) 1581

Open solution

Q12. JEE Main 2026 (Online) 24th January Evening Shift

The largest value of n, for which 40^n divides $60!$, is

(A) 14 (B) 13 (C) 11 (D) 12

Open solution

Q13. JEE Main 2026 (Online) 23rd January Evening Shift

The number of ways, in which 16 oranges can be distributed to four children such that each child gets at least one orange, is

(A) 384 (B) 403 (C) 429 (D) 455

Open solution

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Q14. JEE Main 2026 (Online) 21st January Evening Shift

The largest $n \in \mathbb{N}$, for which 7^n divides $101!$, is :

(A) 18 (B) 15 (C) 19 (D) 16

Open solution

Q15. JEE Main 2026 (Online) 21st January Morning Shift

The number of strictly increasing functions f from the set $\{1,2,3,4,5,6\}$ to the set $\{1,2,3,\ldots,9\}$ such that $f(i) \neq i$ for $1 \leq i \leq 6$, is equal to :

(A) 21 (B) 28 (C) 27 (D) 22

Open solution

Q16. JEE Main 2025 (Online) 8th April Evening Shift

There are 12 points in a plane, no three of which are in the same straight line, except 5 points which are collinear. Then the total number of triangles that can be formed with the vertices at any three of these 12 points is

(A) 230 (B) 210 (C) 200 (D) 220

Open solution

Q17. JEE Main 2025 (Online) 7th April Morning Shift

From a group of 7 batsmen and 6 bowlers, 10 players are to be chosen for a team, which should include at least 4 batsmen and at least 4 bowlers. One batsman and one bowler who are captain and vice-captain respectively of the team should be included. Then the total number of ways such a selection can be made, is

(A) 145 (B) 165 (C) 155 (D) 135

Open solution

Q18. JEE Main 2025 (Online) 3rd April Evening Shift

Line L_1 of slope 2 and line L_2 of slope $\frac{1}{2}$ intersect at the origin O . In the first quadrant, P_1, $P_2, \ldots, P_{12}$ are 12 points on line L_1 and $Q_1, Q_2, \ldots, Q_9$ are 9 points on line L_2. Then the total number of triangles, that can be formed having vertices at three of the 22 points $\mathrm{O}, \mathrm{P}_1, \mathrm{P}_2, \ldots, \mathrm{P}_{12}$, $\mathrm{Q}_1, \mathrm{Q}_2, \ldots, \mathrm{Q}_9$, is:

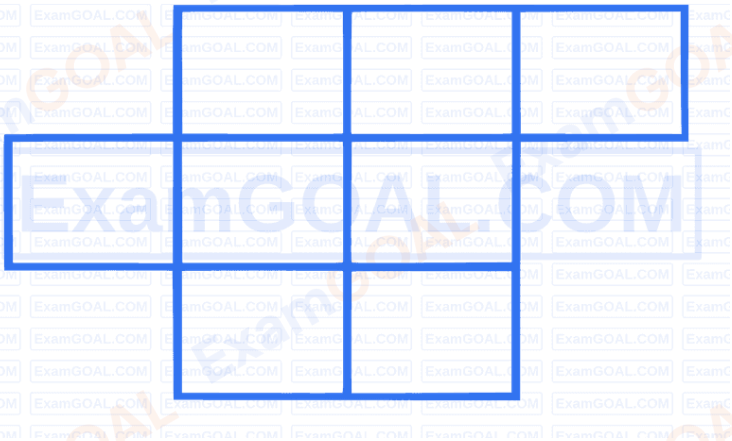
(A) 1026 (B) 1188 (C) 1134 (D) 1080

Open solution

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Q19. JEE Main 2025 (Online) 2nd April Evening Shift

The number of ways, in which the letters A, B, C, D, E can be placed in the 8 boxes of the figure below so that no row remains empty and at most one letter can be placed in a box, is :



Open solution

(A) 5880 (B) 840 (C) 960 (D) 5760

Q20. JEE Main 2025 (Online) 2nd April Morning Shift

The number of sequences of ten terms, whose terms are either 0 or 1 or 2 , that contain exactly five 1 s and exactly three 2 s , is equal to :

(A) 360 (B) 2520 (C) 1820 (D) 45

Open solution

Q21. JEE Main 2025 (Online) 29th January Evening Shift

If all the words with or without meaning made using all the letters of the word "KANPUR" are arranged as in a dictionary, then the word at 440th position in this arrangement is :

(A) PRNAKU (B) PRKAUN (C) PRKANU (D) PRNAUK

Open solution

Q22. JEE Main 2025 (Online) 29th January Morning Shift

Let P be the set of seven digit numbers with sum of their digits equal to 11. If the numbers in P are formed by using the digits 1, 2 and 3 only, then the number of elements in the set P is :

(A) 164 (B) 158 (C) 161 (D) 173

Open solution

Q23. JEE Main 2025 (Online) 28th January Morning Shift

Let ${}^nC_{r-1} = 28$, ${}^nC_r = 56$ and ${}^nC_{r+1} = 70$. Let $A(4\cos t, 4\sin t)$, $B(2\sin t, -2\cos t)$ and $C\left(3r-n, r^2-n-1\right)$ be the vertices of a triangle ABC, where t is a parameter. If $(3x-1)^2 + (3y)^2 = \alpha$, is the locus of the centroid of triangle ABC , then α equals

(A) 18 (B) 8 (C) 20 (D) 6

Open solution

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Q24. JEE Main 2025 (Online) 28th January Morning Shift

The number of different 5 digit numbers greater than 50000 that can be formed using the digits 0 , $1, 2, 3, 4, 5, 6, 7$, such that the sum of their first and last digits should not be more than 8 , is

(A) 5720 (B) 5719 (C) 4608 (D) 4607

Open solution

Q25. JEE Main 2025 (Online) 24th January Evening Shift

Group A consists of 7 boys and 3 girls, while group B consists of 6 boys and 5 girls. The number of ways, 4 boys and 4 girls can be invited for a picnic if 5 of them must be from group A and the remaining 3 from group B, is equal to :

(A) 8925 (B) 9100 (C) 8575 (D) 8750

Open solution

Q26. JEE Main 2025 (Online) 23rd January Morning Shift

The number of words, which can be formed using all the letters of the word "DAUGHTER", so that all the vowels never come together, is :

(A) 34000 (B) 37000 (C) 35000 (D) 36000

Open solution

Q27. JEE Main 2025 (Online) 22nd January Evening Shift

In a group of 3 girls and 4 boys, there are two boys B_1 and B_2. The number of ways, in which these girls and boys can stand in a queue such that all the girls stand together, all the boys stand together, but B_1 and B_2 are not adjacent to each other, is :

(A) 120 (B) 96 (C) 72 (D) 144

Open solution

Q28. JEE Main 2025 (Online) 22nd January Morning Shift

From all the English alphabets, five letters are chosen and are arranged in alphabetical order. The total number of ways, in which the middle letter is ' M ', is :

(A) 6084 (B) 5148 (C) 14950 (D) 4356

Open solution

Q29. JEE Main 2024 (Online) 8th April Evening Shift

The number of ways five alphabets can be chosen from the alphabets of the word MATHEMATICS, where the chosen alphabets are not necessarily distinct, is equal to:

(A) 179 (B) 177 (C) 175 (D) 181

Open solution

Q30. JEE Main 2024 (Online) 6th April Evening Shift

If all the words with or without meaning made using all the letters of the word "NAGPUR" are arranged as in a dictionary, then the word at 315th position in this arrangement is :

(A) NRAPUG (B) NRAGUP (C) NRAPGU (D) NRAGPU

Open solution

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Q31. JEE Main 2024 (Online) 6th April Evening Shift

Let $0 \leq r \leq n$. If ${}^{n+1}C_{r+1} : {}^{n}C_{r} : {}^{n-1}C_{r-1} = 55 : 35 : 21$, then $2n + 5r$ is equal to :

(A) 62 (B) 60 (C) 55 (D) 50

Open solution

Q32. JEE Main 2024 (Online) 6th April Morning Shift

The number of triangles whose vertices are at the vertices of a regular octagon but none of whose sides is a side of the octagon is

(A) 56 (B) 16 (C) 24 (D) 48

Open solution

Q33. JEE Main 2024 (Online) 5th April Evening Shift

Let the set $S = \{2, 4, 8, 16, \ldots, 512\}$ be partitioned into 3 sets A, B, C with equal number of elements such that $\mathrm{A} \cup \mathrm{B} \cup \mathrm{C} = \mathrm{S}$ and $\mathrm{A} \cap \mathrm{B} = \mathrm{B} \cap \mathrm{C} = \mathrm{A} \cap \mathrm{C} = \phi$. The maximum number of such possible partitions of S is equal to:

(A) 1640 (B) 1520 (C) 1710 (D) 1680

Open solution

Q34. JEE Main 2024 (Online) 5th April Evening Shift

60 words can be made using all the letters of the word BHBJO, with or without meaning. If these words are written as in a dictionary, then the 50th word is:

(A) OBBJH (B) HBBJO (C) OBBHJ (D) JBBOH

Open solution

Q35. JEE Main 2024 (Online) 4th April Morning Shift

There are 5 points P_1, P_2, P_3, P_4, P_5 on the side AB, excluding A and B, of a triangle ABC. Similarly there are 6 points $\mathrm{P}_6, \mathrm{P}_7, \ldots, \mathrm{P}_{11}$ on the side BC and 7 points $\mathrm{P}_{12}, \mathrm{P}_{13}, \ldots, \mathrm{P}_{18}$ on the side CA of the triangle. The number of triangles, that can be formed using the points $\mathrm{P}_1, \mathrm{P}_2, \ldots, \mathrm{P}_{18}$ as vertices, is:

(A) 751 (B) 776 (C) 796 (D) 771

Open solution

Q36. JEE Main 2024 (Online) 1st February Morning Shift

If n is the number of ways five different employees can sit into four indistinguishable offices where any office may have any number of persons including zero, then n is equal to :

(A) 47 (B) 53 (C) 51 (D) 43

Open solution

Q37. JEE Main 2024 (Online) 31st January Evening Shift

The number of ways in which 21 identical apples can be distributed among three children such that each child gets at least 2 apples, is

(A) 130 (B) 136 (C) 142 (D) 406

Open solution

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Q38. JEE Main 2024 (Online) 31st January Evening Shift

If for some m, n; ${}^6C_m + 2\left({}^6C_{m+1}\right) + {}^6C_{m+2} > {}^8C_3$ and ${}^{n-1}P_3 : {}^nP_4 = 1 : 8$, then ${}^nP_{m+1} + {}^{n+1}C_m$ is equal to

(A) 380 (B) 376 (C) 372 (D) 384

Open solution

Q39. JEE Main 2024 (Online) 29th January Evening Shift

Number of ways of arranging 8 identical books into 4 identical shelves where any number of shelves may remain empty is equal to

(A) 18 (B) 16 (C) 12 (D) 15

Open solution

Q40. JEE Main 2024 (Online) 27th January Evening Shift

Let $\alpha = \frac{(4!)!}{(4!)^{3!}}$ and $\beta = \frac{(5!)!}{(5!)^{4!}}$. Then :

(A) $\alpha \in \mathbf{N}$ and $\beta \in \mathbf{N}$ (B) $\alpha \in \mathbf{N}$ and $\beta \notin \mathbf{N}$

(C) $\alpha \notin \mathbf{N}$ and $\beta \in \mathbf{N}$ (D) $\alpha \notin \mathbf{N}$ and $\beta \notin \mathbf{N}$

Open solution

Q41. JEE Main 2023 (Online) 15th April Morning Shift

The total number of three-digit numbers, divisible by 3, which can be formed using the digits $1, 3, 5, 8$, if repetition of digits is allowed, is :

(A) 21 (B) 22 (C) 18 (D) 20

Open solution

Q42. JEE Main 2023 (Online) 13th April Evening Shift

All words, with or without meaning, are made using all the letters of the word MONDAY. These words are written as in a dictionary with serial numbers. The serial number of the word MONDAY is :

(A) 324 (B) 328 (C) 326 (D) 327

Open solution

Q43. JEE Main 2023 (Online) 12th April Morning Shift

The number of five digit numbers, greater than 40000 and divisible by 5 , which can be formed using the digits $0, 1, 3, 5, 7$ and 9 without repetition, is equal to :

(A) 132 (B) 72 (C) 120 (D) 96

Open solution

Q44. JEE Main 2023 (Online) 11th April Evening Shift

If the letters of the word MATHS are permuted and all possible words so formed are arranged as in a dictionary with serial numbers, then the serial number of the word THAMS is :

(A) 103 (B) 104 (C) 102 (D) 101

Open solution

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Q45. JEE Main 2023 (Online) 11th April Morning Shift

The number of triplets (x, y, z), where x, y, z are distinct non negative integers satisfying $x + y + z = 15$, is :

(A) 136 (B) 80 (C) 92 (D) 114

Open solution

Q46. JEE Main 2023 (Online) 10th April Evening Shift

Eight persons are to be transported from city A to city B in three cars of different makes. If each car can accommodate at most three persons, then the number of ways, in which they can be transported, is :

(A) 560 (B) 1680 (C) 3360 (D) 1120

Open solution

Q47. JEE Main 2023 (Online) 8th April Evening Shift

If the number of words, with or without meaning, which can be made using all the letters of the word MATHEMATICS in which C and S do not come together, is $(6!)k$, then k is equal to :

(A) 5670 (B) 1890 (C) 2835 (D) 945

Open solution

Q48. JEE Main 2023 (Online) 8th April Morning Shift

The number of arrangements of the letters of the word "INDEPENDENCE" in which all the vowels always occur together is :

(A) 16800 (B) 14800 (C) 18000 (D) 33600

Open solution

Q49. JEE Main 2023 (Online) 8th April Morning Shift

The number of ways, in which 5 girls and 7 boys can be seated at a round table so that no two girls sit together, is :

(A) 720 (B) $7(360)^2$ (C) $7(720)^2$ (D) $126(5!)^2$

Open solution

Q50. JEE Main 2023 (Online) 8th April Morning Shift

Let the number of elements in sets A and B be five and two respectively. Then the number of subsets of $A \times B$ each having at least 3 and at most 6 elements is :

(A) 782 (B) 772 (C) 752 (D) 792

Open solution

Q51. JEE Main 2023 (Online) 6th April Evening Shift

All the letters of the word PUBLIC are written in all possible orders and these words are written as in a dictionary with serial numbers. Then the serial number of the word PUBLIC is :

(A) 578 (B) 576 (C) 580 (D) 582

Open solution

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Q52. JEE Main 2023 (Online) 1st February Morning Shift

The value of $\frac{1}{1!50!}+\frac{1}{3!48!}+\frac{1}{5!46!}+\ldots.+\frac{1}{49!2!}+\frac{1}{51!1!}$ is :

(A) $\frac{2^{51}}{50!}$ (B) $\frac{2^{51}}{51!}$ (C) $\frac{2^{50}}{50!}$ (D) $\frac{2^{50}}{51!}$

Open solution

Q53. JEE Main 2023 (Online) 30th January Evening Shift

The number of ways of selecting two numbers a and $b, a \in \{2,4,6,\ldots.,100\}$ and $b \in \{1,3,5,\ldots..,99\}$ such that 2 is the remainder when $a+b$ is divided by 23 is :

(A) 186 (B) 54 (C) 108 (D) 268

Open solution

Q54. JEE Main 2023 (Online) 29th January Evening Shift

The letters of the word OUGHT are written in all possible ways and these words are arranged as in a dictionary, in a series. Then the serial number of the word TOUGH is :

(A) 79 (B) 84 (C) 89 (D) 86

Open solution

Q55. JEE Main 2023 (Online) 29th January Evening Shift

The number of 3 digit numbers, that are divisible by either 3 or 4 but not divisible by 48, is :

(A) 400 (B) 472 (C) 507 (D) 432

Open solution

Q56. JEE Main 2023 (Online) 25th January Evening Shift

The number of numbers, strictly between 5000 and 10000 can be formed using the digits 1, 3, 5, 7, 9 without repetition, is :

(A) 120 (B) 6 (C) 72 (D) 12

Open solution

Q57. JEE Main 2023 (Online) 24th January Evening Shift

The number of integers, greater than 7000 that can be formed, using the digits 3, 5, 6, 7, 8 without repetition is :

(A) 48 (B) 120 (C) 168 (D) 220

Open solution

Q58. JEE Main 2023 (Online) 24th January Evening Shift

The number of square matrices of order 5 with entries from the set {0, 1}, such that the sum of all the elements in each row is 1 and the sum of all the elements in each column is also 1, is :

(A) 125 (B) 150 (C) 225 (D) 120

Open solution

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Q59. JEE Main 2022 (Online) 30th July Evening Shift

The sum of all values of n, for which ${}^{n-1}C_4 - {}^{n-1}C_3 - \frac{5}{4}\,{}^{n-2}P_2 < 0$, is

(A) 40 (B) 45 (C) 55 (D) 56

Open solution

Q60. JEE Main 2022 (Online) 28th June Evening Shift

The number of ways to distribute 30 identical candies among four children C_1, C_2, C_3 and C_4 so that C_2 receives at least 4 and at most 7 candies, C_3 receives at least 2 and at most 6 candies, is equal to :

(A) 205 (B) 615 (C) 510 (D) 430

Open solution

Q61. JEE Main 2022 (Online) 28th June Morning Shift

The total number of 5-digit numbers, formed by using the digits 1, 2, 3, 5, 6, 7 without repetition, which are multiple of 6, is :

(A) 36 (B) 48 (C) 60 (D) 72

Open solution

Q62. JEE Main 2021 (Online) 1st September Evening Shift

Let P_1, P_2, ..., P_{15} be 15 points on a circle. The number of distinct triangles formed by points P_i, P_j, P_k such that i +j + k $\neq$ 15, is :

(A) 12 (B) 419 (C) 443 (D) 455

Open solution

Q63. JEE Main 2021 (Online) 25th July Evening Shift

If ${}^nP_r = {}^nP_{r+1}$ and ${}^nC_r = {}^nC_{r-1}$, then the value of r is equal to :

(A) 1 (B) 4 (C) 2 (D) 3

Open solution

Q64. JEE Main 2021 (Online) 18th March Morning Shift

The sum of all the 4-digit distinct numbers that can be formed with the digits 1, 2, 2 and 3 is :

(A) 26664 (B) 122664 (C) 122234 (D) 22264

Open solution

Q65. JEE Main 2021 (Online) 17th March Evening Shift

If the sides AB, BC and CA of a triangle ABC have 3, 5 and 6 interior points respectively, then the total number of triangles that can be constructed using these points as vertices, is equal to :

(A) 240 (B) 360 (C) 333 (D) 364

Open solution

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Q66. JEE Main 2021 (Online) 17th March Morning Shift

Team 'A' consists of 7 boys and n girls and Team 'B' has 4 boys and 6 girls. If a total of 52 single matches can be arranged between these two teams when a boy plays against a boy and a girl plays against a girl, then n is equal to :

(A) 5 (B) 2 (C) 4 (D) 6

Open solution

Q67. JEE Main 2021 (Online) 16th March Evening Shift

Consider a rectangle ABCD having 5, 7, 6, 9 points in the interior of the line segments AB, CD, BC, DA respectively. Let α be the number of triangles having these points from different sides as vertices and β be the number of quadrilaterals having these points from different sides as vertices. Then $(\beta - \alpha)$ is equal to :

(A) 717 (B) 795 (C) 1890 (D) 1173

Open solution

Q68. JEE Main 2021 (Online) 26th February Evening Shift

A natural number has prime factorization given by $n = 2^x3^y5^z$, where y and z are such that $y + z = 5$ and $y^{-1} + z^{-1} = \frac{5}{6}$, $y > z$. Then the number of odd divisors of n, including 1, is :

(A) 11 (B) 6 (C) 12 (D) 6x

Open solution

Q69. JEE Main 2021 (Online) 26th February Morning Shift

The number of seven digit integers with sum of the digits equal to 10 and formed by using the digits 1, 2 and 3 only is :

(A) 35 (B) 42 (C) 82 (D) 77

Open solution

Q70. JEE Main 2021 (Online) 25th February Morning Shift

The total number of positive integral solutions (x, y, z) such that xyz = 24 is :

(A) 36 (B) 24 (C) 45 (D) 30

Open solution

Q71. JEE Main 2021 (Online) 24th February Morning Shift

A scientific committee is to be formed from 6 Indians and 8 foreigners, which includes at least 2 Indians and double the number of foreigners as Indians. Then the number of ways, the committee can be formed, is :

(A) 1050 (B) 575 (C) 560 (D) 1625

Open solution

Q72. JEE Main 2020 (Online) 6th September Morning Slot

Two families with three members each and one family with four members are to be seated in a row. In how many ways can they be seated so that the same family members are not separated?

(A) 2! 3! 4! (B) $(3!)^3.(4!)$ (C) $3!\,(4!)^3$ (D) $(3!)^2.(4!)$

Open solution

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Q73. JEE Main 2020 (Online) 5th September Evening Slot

There are 3 sections in a question paper and each section contains 5 questions. A candidate has to answer a total of 5 questions, choosing at least one question from each section. Then the number of ways, in which the candidate can choose the questions, is :

(A) 2250 (B) 2255 (C) 3000 (D) 1500

Open solution

Q74. JEE Main 2020 (Online) 3rd September Morning Slot

The value of $(2.{}^1P_0 - 3.{}^2P_1 + 4.{}^3P_2$ .... up to 51th term)

+ (1! – 2! + 3! – ..... up to 51th term) is equal to :

(A) 1 (B) 1 + (51)! (C) 1 – 51(51)! (D) 1 + (52)!

Open solution

Q75. JEE Main 2020 (Online) 2nd September Evening Slot

Let n > 2 be an integer. Suppose that there are n Metro stations in a city located along a circular path. Each pair of stations is connected by a straight track only. Further, each pair of nearest stations is connected by blue line, whereas all remaining pairs of stations are connected by red line. If the number of red lines is 99 times the number of blue lines, then the value of n is :

(A) 201 (B) 199 (C) 101 (D) 200

Open solution

Q76. JEE Main 2020 (Online) 9th January Morning Slot

If the number of five digit numbers with distinct digits and 2 at the 10th place is 336 k, then k is equal to :

(A) 6 (B) 8 (C) 4 (D) 7

Open solution

Q77. JEE Main 2020 (Online) 8th January Morning Slot

If a, b and c are the greatest value of ${}^{19}C_p$, ${}^{20}C_q$ and ${}^{21}C_r$ respectively, then :

(A) $\frac{a}{11} = \frac{b}{22} = \frac{c}{21}$ (B) $\frac{a}{10} = \frac{b}{22} = \frac{c}{21}$

(C) $\frac{a}{10} = \frac{b}{11} = \frac{c}{42}$ (D) $\frac{a}{11} = \frac{b}{22} = \frac{c}{42}$

Open solution

Q78. JEE Main 2020 (Online) 7th January Evening Slot

The number of ordered pairs (r, k) for which

$6.{}^{35}C_r = (k^2 - 3).{}^{36}C_{r+1}$, where k is an integer, is :

(A) 6 (B) 3 (C) 2 (D) 4

Open solution

Q79. JEE Main 2020 (Online) 7th January Morning Slot

Total number of 6-digit numbers in which only and all the five digits 1, 3, 5, 7 and 9 appear, is :

(A) $\frac{5}{2}(6!)$ (B) $6!$ (C) 5^6 (D) $\frac{1}{2}(6!)$

Open solution

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Q80. JEE Main 2019 (Online) 12th April Evening Slot

A group of students comprises of 5 boys and n girls. If the number of ways, in which a team of 3 students can randomly be selected from this group such that there is at least one boy and at least one girl in each team, is 1750, then n is equal to :

Open solution

(A) 24 (B) 25 (C) 27 (D) 28

Q81. JEE Main 2019 (Online) 12th April Morning Slot

The number of ways of choosing 10 objects out of 31 objects of which 10 are identical and the remaining 21 are distinct, is :

Open solution

(A) $2^{20} - 1$ (B) 2^{20} (C) $2^{20} + 1$ (D) 2^{21}

Q82. JEE Main 2019 (Online) 10th April Evening Slot

Suppose that 20 pillars of the same height have been erected along the boundary of a circular stadium. If the top of each pillar has been connected by beams with the top of all its non-adjacent pillars, then the total number of beams is :

Open solution

(A) 180 (B) 210 (C) 170 (D) 190

Q83. JEE Main 2019 (Online) 10th April Morning Slot

The number of 6 digit numbers that can be formed using the digits 0, 1, 2, 5, 7 and 9 which are divisible by 11 and no digit is repeated is :

Open solution

(A) 36 (B) 60 (C) 72 (D) 48

Q84. JEE Main 2019 (Online) 9th April Morning Slot

A committee of 11 members is to be formed from 8 males and 5 females. If m is the number of ways the committee is formed with at least 6 males and n is the number of ways the committee is formed with at least 3 females, then :

Open solution

(A) n = m – 8 (B) m = n = 78 (C) m + n = 68 (D) m = n = 68

Q85. JEE Main 2019 (Online) 8th April Evening Slot

The number of four-digit numbers strictly greater than 4321 that can be formed using the digits 0,1,2,3,4,5 (repetition of digits is allowed) is :

Open solution

(A) 306 (B) 288 (C) 310 (D) 360

Q86. JEE Main 2019 (Online) 8th April Morning Slot

All possible numbers are formed using the digits 1, 1, 2, 2, 2, 2, 3, 4, 4 taken all at a time. The number of such numbers in which the odd digits occupy even places is :

Open solution

(A) 175 (B) 162 (C) 160 (D) 180

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Q87. JEE Main 2019 (Online) 12th January Evening Slot

There are m men and two women participating in a chess tournament. Each participant plays two games with every other participant. If the number of games played by the men between themselves exceeds the number of games played between the men and the women by 84, then the value of m is :

Open solution

(A) 12 (B) 9 (C) 7 (D) 11

Q88. JEE Main 2019 (Online) 12th January Morning Slot

Consider three boxes, each containing, 10 balls labelled 1, 2, ... , 10. Suppose one ball is randomly drawn from each of the boxes. Denote by n_i, the label of the ball drawn from the ith box, (i = 1, 2, 3). Then, the number of ways in which the balls can be chosen such that $n_1 < n_2 < n_3$ is :

Open solution

(A) 164 (B) 240 (C) 82 (D) 120

Q89. JEE Main 2019 (Online) 9th January Evening Slot

Let S be the set of all triangles in the xy-plane, each having one vertex at the origin and the other two vertices lie on coordinate axes with integral coordinates. If each triangle in S has area 50 sq. units, then the number of elements in the set S is :

Open solution

(A) 9 (B) 18 (C) 36 (D) 32

Q90. JEE Main 2019 (Online) 9th January Evening Slot

The number of natural numbers less than 7,000 which can be formed by using the digits 0, 1, 3, 7, 9 (repitition of digits allowed) is equal to :

Open solution

(A) 374 (B) 372 (C) 375 (D) 250

Q91. JEE Main 2019 (Online) 9th January Morning Slot

Consider a class of 5 girls and 7 boys. The number of different teams consisting of 2 girls and 3 boys that can be formed from this class, if there are two specific boys A and B, who refuse to be the members of the same team, is :

Open solution

(A) 500 (B) 350 (C) 200 (D) 300

Q92. JEE Main 2018 (Online) 16th April Morning Slot

The number of numbers between 2,000 and 5,000 that can be formed with the digits 0, 1, 2, 3, 4 (repetition of digits is not allowed) and are multiple of 3 is :

Open solution

(A) 24 (B) 30 (C) 36 (D) 48

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Q93. JEE Main 2018 (Offline)

From 6 different novels and 3 different dictionaries, 4 novels and 1 dictionary are to be selected and arranged in a row on a shelf so that the dictionary is always in the middle. The number of such arrangements is :

Open solution

(A) at least 750 but less than 1000
(B) at least 1000
(C) less than 500
(D) at least 500 but less than 750

Q94. JEE Main 2018 (Online) 15th April Evening Slot

The number of four letter words that can be formed using the letters of the word **BARRACK** is :

Open solution

(A) 120
(B) 144
(C) 264
(D) 270

Q95. JEE Main 2018 (Online) 15th April Morning Slot

n−digit numbers are formed using only three digits 2, 5 and 7. The smallest value of n for which 900 such distinct numbers can be formed, is :

Open solution

(A) 6
(B) 7
(C) 8
(D) 9

Q96. JEE Main 2017 (Online) 9th April Morning Slot

The number of ways in which 5 boys and 3 girls can be seated on a round table if a particular boy B_1 and a particular girl G_1 never sit adjacent to each other, is :

Open solution

(A) $5 \times 6!$
(B) $6 \times 6!$
(C) $7!$
(D) $5 \times 7!$

Q97. JEE Main 2017 (Online) 8th April Morning Slot

If all the words, with or without meaning, are written using the letters of the word QUEEN and are arranged as in English dictionary, then the position of the word QUEEN is :

Open solution

(A) 44th
(B) 45th
(C) 46th
(D) 47th

Q98. JEE Main 2017 (Offline)

A man X has 7 friends, 4 of them are ladies and 3 are men. His wife Y also has 7 friends, 3 of them are ladies and 4 are men. Assume X and Y have no common friends. Then the total number of ways in which X and Y together can throw a party inviting 3 ladies and 3 men, so that 3 friends of each of X and Y are in this party, is:

Open solution

(A) 468
(B) 469
(C) 484
(D) 485

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Q99. JEE Main 2016 (Online) 10th April Morning Slot

If $\frac{{}^{n+2}C_6}{{}^{n-2}P_2} = 11$, then n satisfies the equation :

(A) $n^2 + 3n - 108 = 0$

(B) $n^2 + 5n - 84 = 0$

(C) $n^2 + 2n - 80 = 0$

(D) $n^2 + n - 110 = 0$

Open solution

Q100. JEE Main 2016 (Online) 10th April Morning Slot

The sum $\sum_{r=1}^{10} (r^2 + 1) \times (r!)$ is equal to :

(A) (11)!

(B) $10 \times (11!)$

(C) $101 \times (10!)$

(D) $11 \times (11!)$

Open solution

Q101. JEE Main 2016 (Online) 9th April Morning Slot

If the four letter words (need not be meaningful) are to be formed using the letters from the word "MEDITERRANEAN" such that the first letter is R and the fourth letter is E, then the total number of all such words is :

(A) $\frac{11!}{(2!)^3}$

(B) 110

(C) 56

(D) 59

Open solution

Q102. JEE Main 2016 (Offline)

If all the words (with or without meaning) having five letters,formed using the letters of the word SMALL and arranged as in a dictionary, then the position of the word SMALL is :

(A) 46th

(B) 59th

(C) 52nd

(D) 58th

Open solution

Q103. JEE Main 2015 (Online) 11th April Morning Slot

Let A = $\{x_1, x_2, \ldots, x_7\}$ and B = $\{y_1, y_2, y_3\}$ be two sets containing seven and three distinct elements respectively. Then the total number of functions f : A $\rightarrow$ B that are onto, if there exist exactly three elements x in A such that $f(x) = y_2$, is equal to :

(A) $14 \cdot {}^7C_2$

(B) $16 \cdot {}^7C_3$

(C) $12 \cdot {}^7C_2$

(D) $14 \cdot {}^7C_3$

Open solution

Q104. JEE Main 2015 (Online) 11th April Morning Slot

If in a regular polygon the number of diagonals is 54, then the number of sides of this polygon is :

(A) 10

(B) 12

(C) 9

(D) 6

Open solution

Q105. JEE Main 2015 (Online) 10th April Morning Slot

The number of ways of selecting 15 teams from 15 men and 15 women, such that each team consists of a man and a woman, is :

(A) 1120

(B) 1240

(C) 1880

(D) 1960

Open solution

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Q106. JEE Main 2015 (Offline)

The number of integers greater than 6,000 that can be formed, using the digits 3, 5, 6, 7 and 8, without repetition, is:

(A) 120 (B) 72 (C) 216 (D) 192

Open solution

Q107. JEE Main 2014 (Online) 19th April Morning Slot

Two women and some men participated in a chess tournament in which every participant played two games with each of the other participants. If the number of games that the men played between themselves exceeds the number of games that the men played with the women by 66, then the number of men who participated in the tournament lies in the interval :

(A) [8, 9] (B) [10, 12] (C) (11, 13) (D) (14, 17)

Open solution

Q108. JEE Main 2014 (Online) 12th April Morning Slot

8 - digit numbers are formed using the digits 1, 1, 2, 2, 2, 3, 4, 4. The number of such numbers in which the odd digits do not occupy odd places, is :

(A) 160 (B) 120 (C) 60 (D) 48

Open solution

Q109. JEE Main 2014 (Online) 11th April Morning Slot

An eight digit number divisible by 9 is to be formed using digits from 0 to 9 without repeating the digits. The number of ways in which this can be done is :

(A) 72 (7!) (B) 18 (7!) (C) 40 (7!) (D) 36 (7!)

Open solution

Q110. JEE Main 2014 (Online) 9th April Morning Slot

The sum of the digits in the unit's place of all the 4-digit numbers formed by using the numbers 3, 4, 5 and 6, without repetition, is :

(A) 432 (B) 108 (C) 36 (D) 18

Open solution

Q111. JEE Main 2013 (Online) 25th April Morning Slot

5 - digit numbers are to be formed using 2, 3, 5, 7, 9 without repeating the digits. If p be the number of such numbers that exceed 20000 and q be the number of those that lie between 30000 and 90000, then p : q is :

(A) 6 : 5 (B) 3 : 2 (C) 4 : 3 (D) 5 : 3

Open solution

Q112. JEE Main 2013 (Online) 23th April Morning Slot

On the sides AB, BC, CA of a $\triangle$ ABC, 3, 4, 5 distinct points (excluding vertices A, B, C) are respectively chosen. The number of triangles that can be constructed using these chosen points as vertices are:

(A) 210 (B) 205 (C) 215 (D) 220

Open solution

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Q113. JEE Main 2013 (Online) 22th April Morning Slot

The number of ways in which an examiner can assign 30 marks to 8 questions, giving not less than 2 marks to any question, is:

(A) ${}^{30}C_7$ (B) ${}^{21}C_8$ (C) ${}^{21}C_7$ (D) ${}^{30}C_8$

Open solution

Q114. JEE Main 2013 (Online) 9th April Morning Slot

A committee of 4 persons is to be formed from 2 ladies, 2 old men and 4 young men such that it includes at least 1 lady, at least 1 old man and at most 2 young men. Then the total number of ways in which this committee can be formed is

(A) 40 (B) 41 (C) 16 (D) 32

Open solution

Q115. JEE Main 2013 (Offline)

Let T_n be the number of all possible triangles formed by joining vertices of an n-sided regular polygon. If $T_{n+1} - T_n$ = 10, then the value of n is :

(A) 7 (B) 5 (C) 10 (D) 8

Open solution

Q116. JEE Main 2013 (Offline)

Let A and B be two sets containing 2 elements and 4 elements respectively. The number of subsets of A $\times$ B having 3 or more elements is :

(A) 219 (B) 211 (C) 256 (D) 220

Open solution

Q117. AIEEE 2012 (Online) 26th May

If seven women and seven men are to be seated around a circular table such that there is a man on either side of every woman, then the number of seating arrangements is

(A) $6!\,7!$ (B) $(6!)^2$ (C) $(7!)^2$ (D) $7!$

Open solution

Q118. AIEEE 2012 (Online) 19th May

The number of arrangements that can be formed from the letters a, b, c, d, e, f taken 3 at a time without repetition and each arrangement containing at least one vowel, is

(A) 96 (B) 128 (C) 24 (D) 72

Open solution

Q119. AIEEE 2012 (Online) 19th May

If $n = {}^mC_2$, then the value of ${}^nC_2$ is given by

(A) $3\left({}^{m+1}C_4\right)$ (B) ${}^{m-1}C_4$ (C) ${}^{m+1}C_4$ (D) $2\left({}^{m+2}C_4\right)$

Open solution

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Q120. AIEEE 2012 (Online) 12th May

Statement 1: If A and B be two sets having p and q elements respectively, where $q > p$. Then the total number of functions from set A to set B is q^p.
Statement 2: The total number of selections of p different objects out of q objects is ${}^qC_p$.

Open solution

(A) Statement 1 is true, Statement 2 is false.

(B) Statement 1 is true, Statement 2 is true; Statement 2 is not a correct explanation of Statement 1.

(C) Statement 1 is false, Statement 2 is true.

(D) Statement 1 is true, Statement 2 is true; Statement 2 is a correct explanation of Statement 1.

Q121. AIEEE 2012 (Online) 7th May

If the number of 5-element subsets of the set $A = \{a_1, a_2, \ldots, a_{20}\}$ of 20 distinct elements is k times the number of 5-element subsets containing a_4, then k is

Open solution

(A) 5 (B) $\frac{20}{7}$ (C) 4 (D) $\frac{10}{3}$

Q122. AIEEE 2012

Assuming the balls to be identical except for difference in colours, the number of ways in which one or more balls can be selected from 10 white, 9 green and 7 black balls is:

Open solution

(A) 880 (B) 629 (C) 630 (D) 879

Q123. AIEEE 2011

These are 10 points in a plane, out of these 6 are collinear, if N is the number of triangles formed by joining these points. then:

Open solution

(A) $N \leq 100$ (B) $100 < N \leq 140$ (C) $140 < N \leq 190$ (D) $N > 190$

Q124. AIEEE 2011

Open solution

Statement - 1: The number of ways of distributing 10 identical balls in 4 distinct boxes such that no box is empty is ${}^9C_3$.
Statement - 2: The number of ways of choosing any 3 places from 9 different places is ${}^9C_3$.

(A) Statement - 1 is true, Statement - 2 is true, Statement - 2 is not a correct explanation for Statement - 1.

(B) Statement - 1 is true, Statement - 2 is false.

(C) Statement - 1 is false, Statement - 2 is true.

(D) Statement - 1 is true, Statement - 2 is true, Statement - 2 is a correct explanation for Statement - 1.

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Q125. AIEEE 2010

There are two urns. Urn A has 3 distinct red balls and urn B has 9 distinct blue balls. From each urn two balls are taken out at random and then transferred to the other. The number of ways in which this can be done is

(A) 36 (B) 66 (C) 108 (D) 3

Open solution

Q126. AIEEE 2009

From 6 different novels and 3 different dictionaries, 4 novels and 1 dictionary are to be selected and arranged in a row on a shelf so that the dictionary is always in the middle. Then the number of such arrangements is :

(A) at least 500 but less than 750

(B) at least 750 but less than 1000

(C) at least 1000

(D) less than 500

Open solution

Q127. AIEEE 2008

In a shop there are five types of ice-cream available. A child buys six ice-cream.

Statement - 1: The number of different ways the child can buy the six ice-cream is $^{10}C_5$.

Statement - 2: The number of different ways the child can buy the six ice-cream is equal to the number of different ways of arranging 6 A and 4 B's in a row.

(A) Statement - 1 is false, Statement - 2 is true

(B) Statement - 1 is true, Statement - 2 is true, Statement - 2 is a correct explanation for Statement - 1

(C) Statement - 1 is true, Statement - 2 is true, Statement - 2 is not a correct explanation for Statement - 1

(D) Statement - 1 is true, Statement - 2 is false

Open solution

Q128. AIEEE 2008

How many different words can be formed by jumbling the letters in the word MISSISSIPPI in which no two S are adjacent?

(A) $8.^6C_4.^7C_4$ (B) $6.7.^8C_4$ (C) $6.8.^7C_4.$ (D) $7.^6C_4.^8C_4$

Open solution

Q129. AIEEE 2007

The set S = {1, 2, 3, ..., 12} is to be partitioned into three sets A, B, C of equal size. Thus $A \cup B \cup C = S$, $A \cap B = B \cap C = A \cap C = \phi$. The number of ways to partition S is

(A) $\frac{12!}{(4!)^3}$ (B) $\frac{12!}{(4!)^4}$ (C) $\frac{12!}{3!\ (4!)^3}$ (D) $\frac{12!}{3!\ (4!)^4}$

Open solution

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Q130. AIEEE 2006

At an election, a voter may vote for any number of candidates, not greater than the number to be elected. There are 10 candidates and 4 are to be selected, if a voter votes for at least one candidate, then the number of ways in which he can vote is

(A) 5040 (B) 6210 (C) 385 (D) 1110

Open solution

Q131. AIEEE 2005

If the letter of the word SACHIN are arranged in all possible ways and these words are written out as in dictionary, then the word SACHIN appears at serial number

(A) 601 (B) 600 (C) 603 (D) 602

Open solution

Q132. AIEEE 2004

How many ways are there to arrange the letters in the word GARDEN with vowels in alphabetical order

(A) 480 (B) 240 (C) 360 (D) 120

Open solution

Q133. AIEEE 2004

The number of ways of distributing 8 identical balls in 3 distinct boxes so that none of the boxes is empty is

(A) ${}^8C_3$ (B) 21 (C) 3^8 (D) 5

Open solution

Q134. AIEEE 2003

A student is to answer 10 out of 13 questions in an examination such that he must choose at least 4 from the first five questions. The number of choices available to him is

(A) 346 (B) 140 (C) 196 (D) 280

Open solution

Q135. AIEEE 2003

The number of ways in which 6 men and 5 women can dine at a round table if no two women are to sit together is given by

(A) $7! \times 5!$ (B) $6! \times 5!$ (C) $30!$ (D) $5! \times 4!$

Open solution

Q136. AIEEE 2003

If ${}^nC_r$ denotes the number of combinations of n things taken r at a time, then the expression ${}^nC_{r+1} + {}^nC_{r-1} + 2 \times {}^nC_r$ equals

(A) ${}^{n+1}C_{r+1}$ (B) ${}^{n+2}C_r$ (C) ${}^{n+2}C_{r+1}$ (D) ${}^{n+1}C_r$

Open solution

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Q137. AIEEE 2002

Total number of four digit odd numbers that can be formed using 0, 1, 2, 3, 5, 7 (using repetition allowed) are :

(A) 216 (B) 375 (C) 400 (D) 720

Open solution

Q138. AIEEE 2002

Numbers greater than 1000 but less than 4000 are formed using the digits 0, 1, 2, 3, 4 (repetition allowed). Their number is :

(A) 125 (B) 105 (C) 374 (D) 625

Open solution

Q139. AIEEE 2002

Five digit numbers divisible by 3 are formed using 0, 1, 2, 3, 4 and 5 without repetition. Total number of such numbers are :

(A) 312 (B) 3125 (C) 120 (D) 216

Open solution

Numerical 94 questions

Q140. JEE Main 2026 (Online) 5th April Morning Shift

Two players A and B play a series of games of badminton. The player, who wins 5 games first, wins the series. Assuming that no game ends in a draw, the number of ways, in which player A wins the series is ____ .

Open solution

Q141. JEE Main 2026 (Online) 28th January Evening Shift

Three persons enter in a lift at the ground floor. The lift will go up to 10th floor. The number of ways, in which the three persons can exit the lift at three different floors, if the lift does not stop at first, second and third floors, is equal to _____.

Open solution

Q142. JEE Main 2026 (Online) 24th January Morning Shift

The number of numbers greater than 5000 , less than 9000 and divisible by 3 , that can be formed using the digits $0, 1, 2, 5, 9$, if the repetition of the digits is allowed, is ____

Open solution

Q143. JEE Main 2026 (Online) 23rd January Evening Shift

Let S denote the set of 4-digit numbers $abcd$ such that $a > b > c > d$ and P denote the set of 5 -digit numbers having product of its digits equal to 20 . Then $n(\mathrm{S}) + n(\mathrm{P})$ is equal to ____

Open solution

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Q144. JEE Main 2026 (Online) 23rd January Morning Shift

The number of 4 -letter words, with or without meaning, which can be formed using the letters PQRPQRSTUVP, is ____ .

Open solution

Q145. JEE Main 2026 (Online) 22nd January Morning Shift

Let ABC be a triangle. Consider four points $\mathrm{p}_1, \mathrm{p}_2, \mathrm{p}_3, \mathrm{p}_4$ on the side AB , five points p_5, p_6, p_7, p_8, p_9 on the side BC, and four points $p_{10}, p_{11}, p_{12}, p_{13}$ on the side AC . None of these points is a vertex of the triangle ABC . Then the total number of pentagons, that can be formed by taking all the vertices from the points $p_1, p_2, \ldots, p_{13}$, is ____

Open solution

Q146. JEE Main 2026 (Online) 21st January Morning Shift

Let $S = \{(m, n) : m, n \in \{1, 2, 3, \ldots\ldots, 50\}\}$. If the number of elements (m, n) in S such that $6^m + 9^n$ is a multiple of 5 is p and the number of elements (m, n) in S such that $m + n$ is a square of a prime number is q , then $\mathrm{p} + \mathrm{q}$ is equal to ____ .

Open solution

Q147. JEE Main 2025 (Online) 4th April Evening Shift

Let m and $\mathrm{n}, (\mathrm{m} < \mathrm{n})$, be two 2-digit numbers. Then the total numbers of pairs (m, n), such that $\gcd(m, n) = 6$, is _____ .

Open solution

Q148. JEE Main 2025 (Online) 3rd April Morning Shift

All five letter words are made using all the letters A, B, C, D, E and arranged as in an English dictionary with serial numbers. Let the word at serial number n be denoted by W_n. Let the probability $\mathrm{P}(\mathrm{W}_\mathrm{n})$ of choosing the word W_n satisfy $\mathrm{P}(\mathrm{W}_\mathrm{n}) = 2\mathrm{P}(\mathrm{W}_{\mathrm{n}-1}), \mathrm{n} > 1$.

If $\mathrm{P}(\mathrm{CDBEA}) = \frac{2^\alpha}{2^\beta - 1}, \alpha, \beta \in \mathbb{N}$, then $\alpha + \beta$ is equal to :_____

Open solution

Q149. JEE Main 2025 (Online) 3rd April Morning Shift

If the number of seven-digit numbers, such that the sum of their digits is even, is $m \cdot n \cdot 10^n; m, n \in \{1, 2, 3, \ldots, 9\}$, then $m + n$ is equal to_____

Open solution

Q150. JEE Main 2025 (Online) 29th January Morning Shift

The number of 6-letter words, with or without meaning, that can be formed using the letters of the word MATHS such that any letter that appears in the word must appear at least twice, is _____.

Open solution

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Q151. JEE Main 2025 (Online) 28th January Evening Shift

The number of natural numbers, between 212 and 999, such that the sum of their digits is 15, is _____.

Open solution

Q152. JEE Main 2025 (Online) 24th January Evening Shift

Number of functions $f : \{1, 2, \ldots, 100\} \to \{0, 1\}$, that assign 1 to exactly one of the positive integers less than or equal to 98 , is equal to _____.

Open solution

Q153. JEE Main 2025 (Online) 24th January Morning Shift

The number of 3 -digit numbers, that are divisible by 2 and 3 , but not divisible by 4 and 9 , is _____.

Open solution

Q154. JEE Main 2025 (Online) 23rd January Evening Shift

The number of ways, 5 boys and 4 girls can sit in a row so that either all the boys sit together or no two boys sit together, is _____.

Open solution

Q155. JEE Main 2024 (Online) 9th April Evening Shift

The number of integers, between 100 and 1000 having the sum of their digits equal to 14 , is _____.

Open solution

Q156. JEE Main 2024 (Online) 8th April Morning Shift

The number of 3-digit numbers, formed using the digits 2, 3, 4, 5 and 7, when the repetition of digits is not allowed, and which are not divisible by 3 , is equal to _____.

Open solution

Q157. JEE Main 2024 (Online) 5th April Morning Shift

The number of ways of getting a sum 16 on throwing a dice four times is _____.

Open solution

Q158. JEE Main 2024 (Online) 4th April Evening Shift

There are 4 men and 5 women in Group A, and 5 men and 4 women in Group B. If 4 persons are selected from each group, then the number of ways of selecting 4 men and 4 women is _____.

Open solution

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Q159. JEE Main 2024 (Online) 1st February Morning Shift

The number of elements in the set $S = \{(x, y, z) : x, y, z \in \mathbf{Z}, x + 2y + 3z = 42, x, y, z \geqslant 0\}$ equals _____.

Open solution

Q160. JEE Main 2024 (Online) 31st January Morning Shift

The total number of words (with or without meaning) that can be formed out of the letters of the word '**DISTRIBUTION**' taken four at a time, is equal to _____.

Open solution

Q161. JEE Main 2024 (Online) 30th January Evening Shift

In an examination of Mathematics paper, there are 20 questions of equal marks and the question paper is divided into three sections : A, B and C. A student is required to attempt total 15 questions taking at least 4 questions from each section. If section A has 8 questions, section B has 6 questions and section C has 6 questions, then the total number of ways a student can select 15 questions is _____.

Open solution

Q162. JEE Main 2024 (Online) 29th January Morning Shift

All the letters of the word "GTWENTY" are written in all possible ways with or without meaning and these words are written as in a dictionary. The serial number of the word "GTWENTY" is _____.

Open solution

Q163. JEE Main 2023 (Online) 15th April Morning Shift

A person forgets his 4-digit ATM pin code. But he remembers that in the code all the digits are different, the greatest digit is 7 and the sum of the first two digits is equal to the sum of the last two digits. Then the maximum number of trials necessary to obtain the correct code is _____.

Open solution

Q164. JEE Main 2023 (Online) 13th April Evening Shift

Total numbers of 3-digit numbers that are divisible by 6 and can be formed by using the digits $1, 2, 3, 4, 5$ with repetition, is _____.

Open solution

Q165. JEE Main 2023 (Online) 13th April Morning Shift

The number of seven digit positive integers formed using the digits $1, 2, 3$ and 4 only and sum of the digits equal to 12 is _____.

Open solution

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Q166. JEE Main 2023 (Online) 12th April Morning Shift

Let the digits a, b, c be in A. P. Nine-digit numbers are to be formed using each of these three digits thrice such that three consecutive digits are in A.P. at least once. How many such numbers can be formed?

Open solution

Q167. JEE Main 2023 (Online) 11th April Morning Shift

In an examination, 5 students have been allotted their seats as per their roll numbers. The number of ways, in which none of the students sits on the allotted seat, is _____.

Open solution

Q168. JEE Main 2023 (Online) 10th April Evening Shift

The sum of all the four-digit numbers that can be formed using all the digits 2, 1, 2, 3 is equal to _____.

Open solution

Q169. JEE Main 2023 (Online) 10th April Morning Shift

The number of permutations, of the digits 1, 2, 3, ..., 7 without repetition, which neither contain the string 153 nor the string 2467, is _____.

Open solution

Q170. JEE Main 2023 (Online) 10th April Morning Shift

Some couples participated in a mixed doubles badminton tournament. If the number of matches played, so that no couple played in a match, is 840, then the total number of persons, who participated in the tournament, is _____.

Open solution

Q171. JEE Main 2023 (Online) 6th April Evening Shift

The number of 4-letter words, with or without meaning, each consisting of 2 vowels and 2 consonants, which can be formed from the letters of the word UNIVERSE without repetition is _____.

Open solution

Q172. JEE Main 2023 (Online) 6th April Morning Shift

The number of ways of giving 20 distinct oranges to 3 children such that each child gets at least one orange is _____.

Open solution

Q173. JEE Main 2023 (Online) 1st February Evening Shift

Number of integral solutions to the equation $x + y + z = 21$, where $x \geq 1, y \geq 3, z \geq 4$, is equal to _____.

Open solution

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Q174. JEE Main 2023 (Online) 1st February Evening Shift

The total number of six digit numbers, formed using the digits 4, 5, 9 only and divisible by 6, is _____.

Open solution

Q175. JEE Main 2023 (Online) 1st February Morning Shift

The number of 3-digit numbers, that are divisible by either 2 or 3 but not divisible by 7, is _____.

Open solution

Q176. JEE Main 2023 (Online) 1st February Morning Shift

The number of words, with or without meaning, that can be formed using all the letters of the word ASSASSINATION so that the vowels occur together, is _____.

Open solution

Q177. JEE Main 2023 (Online) 31st January Evening Shift

Let $A = [a_{ij}], a_{ij} \in \mathbb{Z} \cap [0, 4], 1 \leq i, j \leq 2$.

The number of matrices A such that the sum of all entries is a prime number $p \in (2, 13)$ is _____.

Open solution

Q178. JEE Main 2023 (Online) 31st January Evening Shift

If ${}^{2n+1}P_{n-1} : {}^{2n-1}P_n = 11 : 21$,

then $n^2 + n + 15$ is equal to :

Open solution

Q179. JEE Main 2023 (Online) 31st January Morning Shift

Let 5 digit numbers be constructed using the digits $0, 2, 3, 4, 7, 9$ with repetition allowed, and are arranged in ascending order with serial numbers. Then the serial number of the number 42923 is _____.

Open solution

Q180. JEE Main 2023 (Online) 31st January Morning Shift

Number of 4-digit numbers that are less than or equal to 2800 and either divisible by 3 or by 11 , is equal to _____.

Open solution

Q181. JEE Main 2023 (Online) 30th January Evening Shift

The number of seven digits odd numbers, that can be formed using all the

seven digits 1, 2, 2, 2, 3, 3, 5 is _____.

Open solution

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Q182. JEE Main 2023 (Online) 30th January Morning Shift

Number of 4-digit numbers (the repetition of digits is allowed) which are made using the digits 1, 2, 3 and 5, and are divisible by 15, is equal to _____.

Open solution

Q183. JEE Main 2023 (Online) 29th January Evening Shift

The total number of 4-digit numbers whose greatest common divisor with 54 is 2, is _____.

Open solution

Q184. JEE Main 2023 (Online) 29th January Morning Shift

If all the six digit numbers $x_1\,x_2\,x_3\,x_4\,x_5\,x_6$ with $0 < x_1 < x_2 < x_3 < x_4 < x_5 < x_6$ are arranged in the increasing order, then the sum of the digits in the 72nd number is _____.

Open solution

Q185. JEE Main 2023 (Online) 29th January Morning Shift

Five digit numbers are formed using the digits 1, 2, 3, 5, 7 with repetitions and are written in descending order with serial numbers. For example, the number 77777 has serial number 1. Then the serial number of 35337 is _____.

Open solution

Q186. JEE Main 2023 (Online) 25th January Evening Shift

A triangle is formed by X-axis, Y-axis and the line $3x + 4y = 60$. Then the number of points P(a, b) which lie strictly inside the triangle, where a is an integer and b is a multiple of a, is _____.

Open solution

Q187. JEE Main 2023 (Online) 25th January Evening Shift

Suppose Anil's mother wants to give 5 whole fruits to Anil from a basket of 7 red apples, 5 white apples and 8 oranges. If in the selected 5 fruits, at least 2 oranges, at least one red apple and at least one white apple must be given, then the number of ways, Anil's mother can offer 5 fruits to Anil is _____

Open solution

Q188. JEE Main 2023 (Online) 25th January Morning Shift

Let x and y be distinct integers where $1 \le x \le 25$ and $1 \le y \le 25$. Then, the number of ways of choosing x and y, such that $x + y$ is divisible by 5, is _____.

Open solution

Q189. JEE Main 2023 (Online) 24th January Morning Shift

A boy needs to select five courses from 12 available courses, out of which 5 courses are language courses. If he can choose at most two language courses, then the number of ways he can choose five courses is _____

Open solution

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Q190. JEE Main 2023 (Online) 24th January Morning Shift

The number of 9 digit numbers, that can be formed using all the digits of the number 123412341 so that the even digits occupy only even places, is _____.

Open solution

Q191. JEE Main 2022 (Online) 29th July Evening Shift

The number of natural numbers lying between 1012 and 23421 that can be formed using the digits $2, 3, 4, 5, 6$ (repetition of digits is not allowed) and divisible by 55 is _____.

Open solution

Q192. JEE Main 2022 (Online) 29th July Morning Shift

The number of matrices of order 3×3, whose entries are either 0 or 1 and the sum of all the entries is a prime number, is _____.

Open solution

Q193. JEE Main 2022 (Online) 28th July Evening Shift

A class contains b boys and g girls. If the number of ways of selecting 3 boys and 2 girls from the class is 168 , then $\mathrm{b} + 3\,\mathrm{g}$ is equal to _____.

Open solution

Q194. JEE Main 2022 (Online) 28th July Morning Shift

Let S be the set of all passwords which are six to eight characters long, where each character is either an alphabet from $\{A, B, C, D, E\}$ or a number from $\{1, 2, 3, 4, 5\}$ with the repetition of characters allowed. If the number of passwords in S whose at least one character is a number from $\{1, 2, 3, 4, 5\}$ is $\alpha \times 5^6$, then α is equal to _____.

Open solution

Q195. JEE Main 2022 (Online) 26th July Evening Shift

Numbers are to be formed between 1000 and 3000 , which are divisible by 4 , using the digits $1, 2, 3, 4, 5$ and 6 without repetition of digits. Then the total number of such numbers is _____.

Open solution

Q196. JEE Main 2022 (Online) 26th July Morning Shift

The number of 5-digit natural numbers, such that the product of their digits is 36 , is _____.

Open solution

Q197. JEE Main 2022 (Online) 25th July Morning Shift

The letters of the word 'MANKIND' are written in all possible orders and arranged in serial order as in an English dictionary. Then the serial number of the word 'MANKIND' is _____.

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Q198. JEE Main 2022 (Online) 30th June Morning Shift

The number of 6-digit numbers made by using the digits 1, 2, 3, 4, 5, 6, 7, without repetition and which are multiple of 15 is _____.

Open solution

Q199. JEE Main 2022 (Online) 29th June Evening Shift

The total number of four digit numbers such that each of first three digits is divisible by the last digit, is equal to _____.

Open solution

Q200. JEE Main 2022 (Online) 29th June Morning Shift

Let $b_1b_2b_3b_4$ be a 4-element permutation with $b_i \in \{1, 2, 3, ..., 100\}$ for $1 \leq i \leq 4$ and $b_i \neq b_j$ for $i \neq j$, such that either b_1, b_2, b_3 are consecutive integers or b_2, b_3, b_4 are consecutive integers. Then the number of such permutations $b_1b_2b_3b_4$ is equal to _____.

Open solution

Q201. JEE Main 2022 (Online) 27th June Evening Shift

Let A be a matrix of order 2×2, whose entries are from the set {0, 1, 2, 3, 4, 5}. If the sum of all the entries of A is a prime number p, $2 < p < 8$, then the number of such matrices A is _____.

Open solution

Q202. JEE Main 2022 (Online) 27th June Morning Shift

The number of ways, 16 identical cubes, of which 11 are blue and rest are red, can be placed in a row so that between any two red cubes there should be at least 2 blue cubes, is _____.

Open solution

Q203. JEE Main 2022 (Online) 26th June Evening Shift

The total number of 3-digit numbers, whose greatest common divisor with 36 is 2, is _____.

Open solution

Q204. JEE Main 2022 (Online) 26th June Morning Shift

There are ten boys $B_1, B_2, ..., B_{10}$ and five girls $G_1, G_2, ..., G_5$ in a class. Then the number of ways of forming a group consisting of three boys and three girls, if both B_1 and B_2 together should not be the members of a group, is _____.

Open solution

Q205. JEE Main 2022 (Online) 25th June Evening Shift

The total number of three-digit numbers, with one digit repeated exactly two times, is _____.

Open solution

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Q206. JEE Main 2022 (Online) 25th June Morning Shift

The number of 3-digit odd numbers, whose sum of digits is a multiple of 7, is _____.

Open solution

Q207. JEE Main 2022 (Online) 25th June Morning Shift

Let A be a 3×3 matrix having entries from the set $\{-1, 0, 1\}$. The number of all such matrices A having sum of all the entries equal to 5, is _____.

Open solution

Q208. JEE Main 2022 (Online) 24th June Evening Shift

The number of 7-digit numbers which are multiples of 11 and are formed using all the digits 1, 2, 3, 4, 5, 7 and 9 is _____.

Open solution

Q209. JEE Main 2022 (Online) 24th June Morning Shift

In an examination, there are 5 multiple choice questions with 3 choices, out of which exactly one is correct. There are 3 marks for each correct answer, -2 marks for each wrong answer and 0 mark if the question is not attempted. Then, the number of ways a student appearing in the examination gets 5 marks is _____.

Open solution

Q210. JEE Main 2021 (Online) 1st September Evening Shift

All the arrangements, with or without meaning, of the word FARMER are written excluding any word that has two R appearing together. The arrangements are listed serially in the alphabetic order as in the English dictionary. Then the serial number of the word FARMER in this list is _____.

Open solution

Q211. JEE Main 2021 (Online) 31st August Evening Shift

The number of 4-digit numbers which are neither multiple of 7 nor multiple of 3 is _____.

Open solution

Q212. JEE Main 2021 (Online) 31st August Morning Shift

The number of six letter words (with or without meaning), formed using all the letters of the word 'VOWELS', so that all the consonants never come together, is _____.

Open solution

Q213. JEE Main 2021 (Online) 27th August Evening Shift

Let S = {1, 2, 3, 4, 5, 6, 9}. Then the number of elements in the set T = $\{A \subseteq S : A \neq \phi$ and the sum of all the elements of A is not a multiple of 3$\}$ is _____.

Open solution

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Q214. JEE Main 2021 (Online) 27th August Morning Shift

A number is called a palindrome if it reads the same backward as well as forward. For example 285582 is a six digit palindrome. The number of six digit palindromes, which are divisible by 55, is _____.

Open solution

Q215. JEE Main 2021 (Online) 26th August Morning Shift

If ${}^{1}P_1 + 2.{}^{2}P_2 + 3.{}^{3}P_3 + \ldots + 15.{}^{15}P_{15} = {}^{q}P_r - s, 0 \le s \le 1$, then ${}^{q+s}C_{r-s}$ is equal to _____.

Open solution

Q216. JEE Main 2021 (Online) 26th August Morning Shift

The number of three-digit even numbers, formed by the digits 0, 1, 3, 4, 6, 7 if the repetition of digits is not allowed, is _____.

Open solution

Q217. JEE Main 2021 (Online) 27th July Evening Shift

Let n be a non-negative integer. Then the number of divisors of the form "4n + 1" of the number $(10)^{10} \cdot (11)^{11} \cdot (13)^{13}$ is equal to _____.

Open solution

Q218. JEE Main 2021 (Online) 25th July Morning Shift

There are 5 students in class 10, 6 students in class 11 and 8 students in class 12. If the number of ways, in which 10 students can be selected from them so as to include at least 2 students from each class and at most 5 students from the total 11 students of class 10 and 11 is 100 k, then k is equal to _____.

Open solution

Q219. JEE Main 2021 (Online) 22th July Evening Shift

If the digits are not allowed to repeat in any number formed by using the digits 0, 2, 4, 6, 8, then the number of all numbers greater than 10,000 is equal to _____.

Open solution

Q220. JEE Main 2021 (Online) 20th July Morning Shift

There are 15 players in a cricket team, out of which 6 are bowlers, 7 are batsman and 2 are wicketkeepers. The number of ways, a team of 11 players be selected from them so as to include at least 4 bowlers, 5 batsman and 1 wicketkeeper, is _____.

Open solution

Q221. JEE Main 2021 (Online) 18th March Evening Shift

If $\sum_{r=1}^{10} r!(r^3 + 6r^2 + 2r + 5) = \alpha(11!)$, then the value of α is equal to _____.

Open solution

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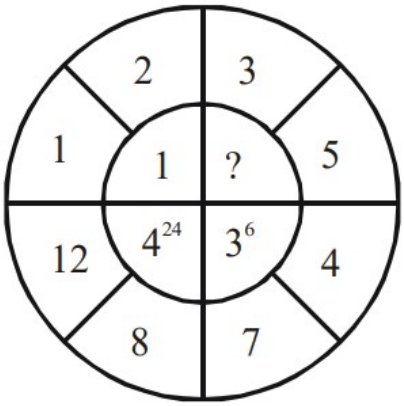
Q222. JEE Main 2021 (Online) 18th March Morning Shift

The number of times the digit 3 will be written when listing the integers from 1 to 1000 is :

Open solution

Q223. JEE Main 2021 (Online) 18th March Morning Shift

The missing value in the following figure is



Open solution

Q224. JEE Main 2021 (Online) 26th February Evening Shift

The total number of 4-digit numbers whose greatest common divisor with 18 is 3, is _____.

Open solution

Q225. JEE Main 2021 (Online) 25th February Morning Shift

The total number of numbers, lying between 100 and 1000 that can be formed with the digits 1, 2, 3, 4, 5, if the repetition of digits is not allowed and numbers are divisible by either 3 or 5, is _____.

Open solution

Q226. JEE Main 2021 (Online) 24th February Evening Shift

The students S_1, S_2, ..., S_{10} are to be divided into 3 groups A, B and C such that each group has at least one student and the group C has at most 3 students. Then the total number of possibilities of forming such groups is _____.

Open solution

Q227. JEE Main 2020 (Online) 6th September Evening Slot

The number of words (with or without meaning) that can be formed from all the letters of the word "LETTER" in which vowels never come together is _____ .

Open solution

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Q228. JEE Main 2020 (Online) 5th September Morning Slot

The number of words, with or without meaning, that can be formed by taking 4 letters at a time from the letters of the word 'SYLLABUS' such that two letters are distinct and two letters are alike, is :

Open solution

Q229. JEE Main 2020 (Online) 4th September Evening Slot

A test consists of 6 multiple choice questions, each having 4 alternative answers of which only one is correct. The number of ways, in which a candidate answers all six questions such that exactly four of the answers are correct, is _____.

Open solution

Q230. JEE Main 2020 (Online) 3rd September Evening Slot

The total number of 3-digit numbers, whose sum of digits is 10, is _____.

Open solution

Q231. JEE Main 2020 (Online) 2nd September Morning Slot

If the letters of the word 'MOTHER' be permuted and all the words so formed (with or without meaning) be listed as in a dictionary, then the position of the word 'MOTHER' is _____.

Open solution

Q232. JEE Main 2020 (Online) 8th January Evening Slot

The number of 4 letter words (with or without meaning) that can be formed from the eleven letters of the word 'EXAMINATION' is _____.

Open solution

Q233. JEE Main 2020 (Online) 8th January Morning Slot

An urn contains 5 red marbles, 4 black marbles and 3 white marbles. Then the number of ways in which 4 marbles can be drawn so that at the most three of them are red is _____.

Open solution

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Answer Key

MCQ (Single Correct Answer)

1. **B**	2. **C**	3. **D**	4. **A**	5. **C**
6. **D**	7. **C**	8. **C**	9. **B**	10. **B**
11. **C**	12. **A**	13. **D**	14. **D**	15. **B**
16. **B**	17. **C**	18. **C**	19. **D**	20. **B**
21. **B**	22. **C**	23. **C**	24. **D**	25. **A**
26. **D**	27. **D**	28. **B**	29. **A**	30. **C**
31. **D**	32. **B**	33. **D**	34. **A**	35. **A**
36. **C**	37. **B**	38. **C**	39. **D**	40. **A**
41. **B**	42. **D**	43. **C**	44. **A**	45. **D**
46. **B**	47. **A**	48. **A**	49. **D**	50. **D**
51. **D**	52. **D**	53. **C**	54. **C**	55. **D**
56. **C**	57. **C**	58. **D**	59. **B**	60. **D**
61. **D**	62. **C**	63. **C**	64. **A**	65. **C**
66. **C**	67. **A**	68. **C**	69. **D**	70. **D**
71. **D**	72. **B**	73. **A**	74. **D**	75. **A**
76. **B**	77. **D**	78. **D**	79. **A**	80. **B**
81. **B**	82. **C**	83. **B**	84. **B**	85. **C**
86. **D**	87. **A**	88. **D**	89. **C**	90. **A**
91. **D**	92. **B**	93. **B**	94. **D**	95. **B**
96. **A**	97. **C**	98. **D**	99. **A**	100. **B**
101. **D**	102. **D**	103. **D**	104. **B**	105. *Scan QR*
106. **D**	107. **B**	108. **B**	109. **D**	110. **B**
111. **D**	112. **B**	113. **C**	114. **B**	115. **B**
116. **A**	117. **A**	118. **A**	119. **A**	120. **D**
121. **C**	122. **D**	123. **A**	124. **A**	125. **C**
126. **C**	127. **A**	128. **D**	129. **A**	130. **C**
131. **A**	132. **C**	133. **B**	134. **C**	135. **B**
136. **C**	137. **D**	138. **C**	139. **D**	

Numerical

140. **126**	141. **210**	142. **42**	143. **260**	144. **1422**

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145. **660**	146. **1333**	147. **64**	148. **183**	149. **14**
150. **1405**	151. **64**	152. **392**	153. **125**	154. **17280**
155. **70**	156. **36**	157. **125**	158. **5626**	159. **169**
160. **3734**	161. **11376**	162. **553**	163. **72**	164. **16**
165. **413**	166. **1260**	167. **44**	168. **26664**	169. **4898**
170. **16**	171. **432**	172. **3483638676**	173. **105**	174. **81**
175. **514**	176. **50400**	177. **204**	178. **45**	179. **2997**
180. **710**	181. **240**	182. **21**	183. **3000**	184. **32**
185. **1436**	186. **31**	187. **6860 or 3**	188. **120**	189. **546**
190. **60**	191. **6**	192. **282**	193. **17**	194. **7073**
195. **30**	196. **180**	197. **1492**	198. **360**	199. **1086**
200. **18915**	201. **180**	202. **56**	203. **150**	204. **1120**
205. **243**	206. **63**	207. **414**	208. **576**	209. **40**
210. **77**	211. **5143**	212. **576**	213. **80**	214. **100**
215. **136**	216. **52**	217. **924**	218. **238**	219. **96**
220. **777**	221. **160**	222. **300**	223. **4**	224. **1000**
225. **32**	226. **31650**	227. **120**	228. **240**	229. **135**
230. **54**	231. **309**	232. **2454**	233. **490**	

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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